BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Passwords, Encryption and Penetration Testing

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How does a firewall protect a network?

2. Which is an IPv6 address?
   A) 2001:db8::7
   B) 192.168.1.7
   C) 02:11:22:33:44:55
   D) school.example

3. What is a MAC address used for and how is it usually written?

4. Use decimal units. A file is 3 MB. Give its size in KB and bytes.

Today's objectives

Explain how authentication and account lockout reduce unauthorised access. Explain how encryption protects intercepted data.

Key vocabulary - Read, cover, recall

Authentication
Checking that someone is who they claim to be.

Strong password
Long and difficult-to-guess unique secret.

Two-factor authentication
Using two different forms of evidence to sign in.

Account lockout
Temporarily blocking access after repeated failed attempts.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Strong passwords and authentication

Long, unique passwords make guessing harder. Multi-factor or two-step authentication adds a separate proof of identity.

Account lockout

Limiting repeated incorrect login attempts reduces opportunities for brute-force guessing.

Encryption

Encryption converts readable plaintext into ciphertext using an algorithm and key. Captured data is not readily understandable without the correct key.

Penetration testing

An authorised test looks for weaknesses so they can be fixed before attackers exploit them. It must be carried out with explicit permission and an agreed scope.

I Do - Worked model

A school wants to reduce the chance of attackers signing in with guessed passwords. Explain how strong passwords and account lockout help. [4 marks]

- The risk is many attempted password guesses.
- Long unique passwords increase the number of possibilities [1], making guessing the correct password harder [1].
- A lockout policy blocks further attempts after repeated failures [1], reducing how many guesses an attacker can try [1].
- Two protections, each explained with how the attacker is limited.

We Do - Guided practice

1. A staff member accesses sensitive files away from school. Explain why two-factor authentication is useful. [4 marks]

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2. A website sends account information across the Internet. Explain why encryption is important. [4 marks]

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3. A college hires authorised testers before launching an online system. Explain the purpose and two benefits of penetration testing. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the account response.

```
failed = 4
if failed >= 3 then
print("Lock account")
endif
```

You Do - Independent exam questions

Q2. State what encryption does to data. [2 marks]

Q3. An online sports club membership portal is targeted by repeated password guesses. Explain how account lockout and two-factor authentication can help protect member accounts. [4 marks]

Q4. Explain how strong passwords, encryption and penetration testing improve network security. [6 marks]

Exit Ticket - True or False?

Penetration testing should be conducted only with permission from the system owner.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It uses rules to allow or block network traffic.

2. A - 2001:db8::7

3. A MAC address identifies a network interface on a LAN. It is usually six pairs of hexadecimal digits.

4. (a) 3,000 KB and 3,000,000 bytes.

I Do - Explain the model

The risk is many attempted password guesses.

Long unique passwords increase the number of possibilities [1], making guessing the correct password harder [1].

A lockout policy blocks further attempts after repeated failures [1], reducing how many guesses an attacker can try [1].

Two protections, each explained with how the attacker is limited.

We Do - Model answers

1. The login needs an additional independent authentication factor [1], such as a code on a separate device [1]. A stolen password alone is therefore insufficient [1], reducing unauthorised access [1].

2. The data is transformed from readable plaintext [1] into scrambled ciphertext [1]. If transmission is intercepted [1], attackers without the appropriate key cannot readily read the original information [1].

3. Authorised testers examine the system for vulnerabilities [1], using a defined scope [1]. They report weaknesses [1], allowing the organisation to correct them before malicious attackers exploit them [1].

Code First - Answer

```
failed = 4
if failed >= 3 then
print("Lock account")
endif
```

Expected output: Lock account

You Do - Self-mark

Q2: Transforms readable plaintext [1] into unreadable ciphertext without the correct key [1].

Q3: Account lockout limits further login attempts after repeated incorrect passwords [1], making brute-force guessing less effective [1]. Two-factor authentication asks for a second independent proof [1], so guessing a password alone is insufficient to enter the member's account [1].

Q4: Strong passwords are difficult to guess [1], reducing unauthorised logins [1]. Encryption scrambles transmitted data [1], protecting contents if intercepted [1]. Authorised penetration testing identifies vulnerabilities [1], enabling patches/improvements [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.